INTRO TO QGIS

for Emergency Communications Mapping

Ham Radio EmComm · CERT · Situational Awareness

Rick Stoner - KC3DRU

October 2026

https://github.com/KC3DRU/EmComm-Mapping-Demo

What We'll Cover

01 **Why maps matter in EmComm**
Situational awareness, after-action, planning

02 **Why QGIS (and when not)**
Open-source power vs specialized radio tools

03 **Core GIS concepts in 5 minutes**
Layers, vectors, CRS, attributes

04 **Hands-on workflow**
Points, lines, labels, signal-strength styling

05 **Data, offline use & next steps**
Sources, printing, resources

Why Maps for EmComm?

Who can talk to whom

Visualize net connectivity, simplex paths, and relay needs during an activation.

Where people & assets are

Stations, shelters, EOCs, repeaters, staging areas — all on one picture.

After-action & planning

Document what worked, coverage gaps, and plan future exercises or deployments.

How to Show Your Contacts?

What types of things would you want to show on a map to visualize who you were able to make contact with during a net?

How to Show Your Contacts?

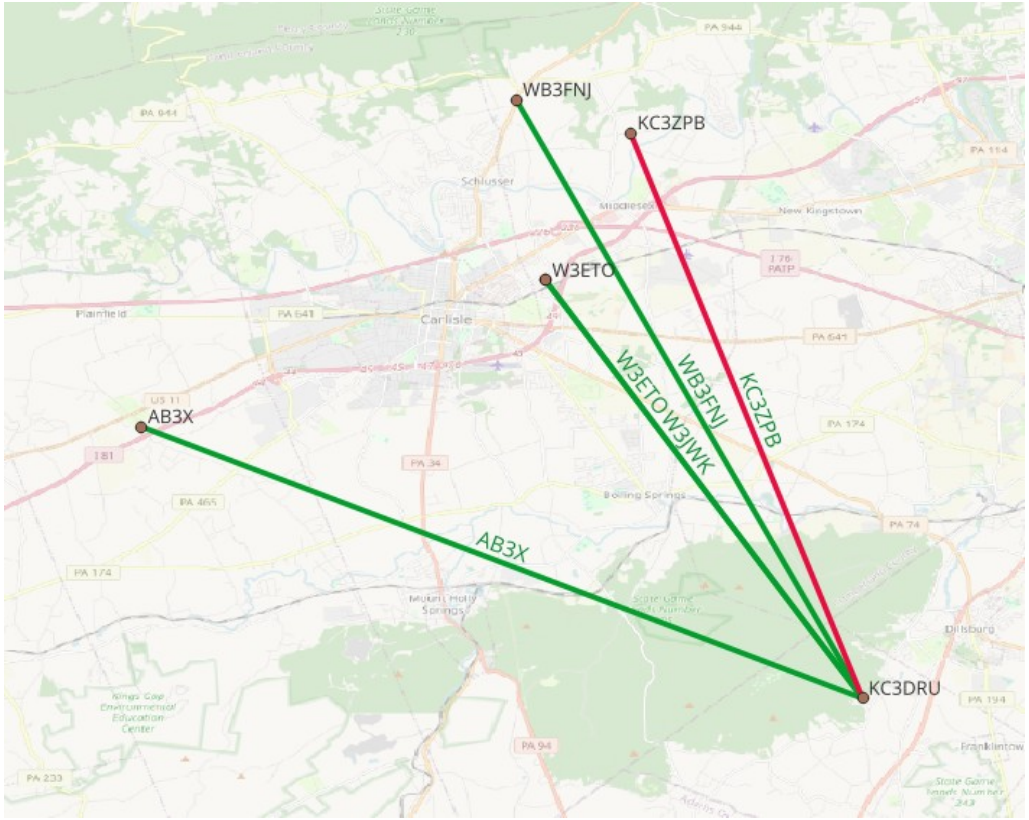
Simple map features that can show your network:

Points	Each station / operator location, labeled with callsign
Lines	Radio contacts between stations
Color	Line color = perceived signal strength (strong / fair / weak)
Result	Instant picture of net topology and where relays were needed

July 2026 Simplex Test

Operators that KC3DRU was able to hear over simplex

Map made with QGIS Software



What is QGIS?

Free and open-source Geographic Information System (GIS) desktop software

Cost	Completely free — no license fees
Platforms	Windows, macOS, Linux, even Raspberry Pi
Community	Huge user base + plugins for almost everything
Capability	From simple pins to advanced analysis & print layouts
Data	Reads shapefiles, GeoJSON, GPX, CSV, raster, WMS, etc.

Why QGIS — and When Not

QGIS is excellent for…

- Custom situational maps (any layers you want)
- After-action reports & printed field maps
- Combining radio data + public GIS + OSM
- Rich symbology, labeling, print layout
- Multiple base maps to help orient viewer
- Data exportable to other apps

Other, Specialized tools shine for…

- YAAC / Xastir / APRSISCE — live APRS tracking
- Graywolf / similar — radio-specific mapping
- CalTopo – mobile access, team shared maps
- When you need real-time position reports
- QGIS is complementary, not a replacement

Core Concepts

Base Map

Start with a map on the bottom of the stack. The base map will change depending on project.

Vector data

Points (stations), lines (paths/contacts), polygons (areas). Attributes live in a table.

Layers

Think of transparent sheets stacked. Basemap + stations + contacts + labels.

Symbology

How features look: color, size, line style — driven by attribute values (e.g., signal strength).

CRS / Projection

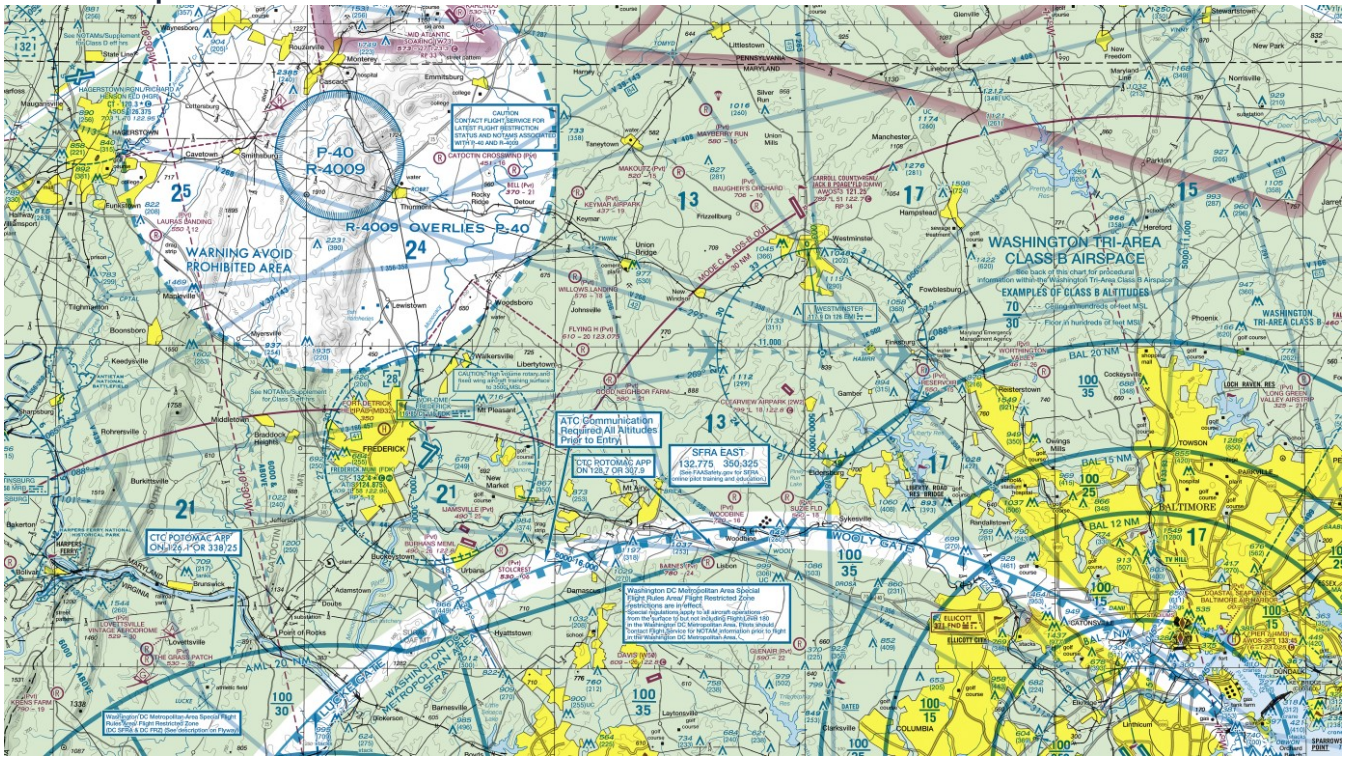
How the round Earth is flattened. Use a local projected CRS or Web Mercator for web maps.

Labels

Text pulled from attributes (callsign, name, grid square) placed on the map.

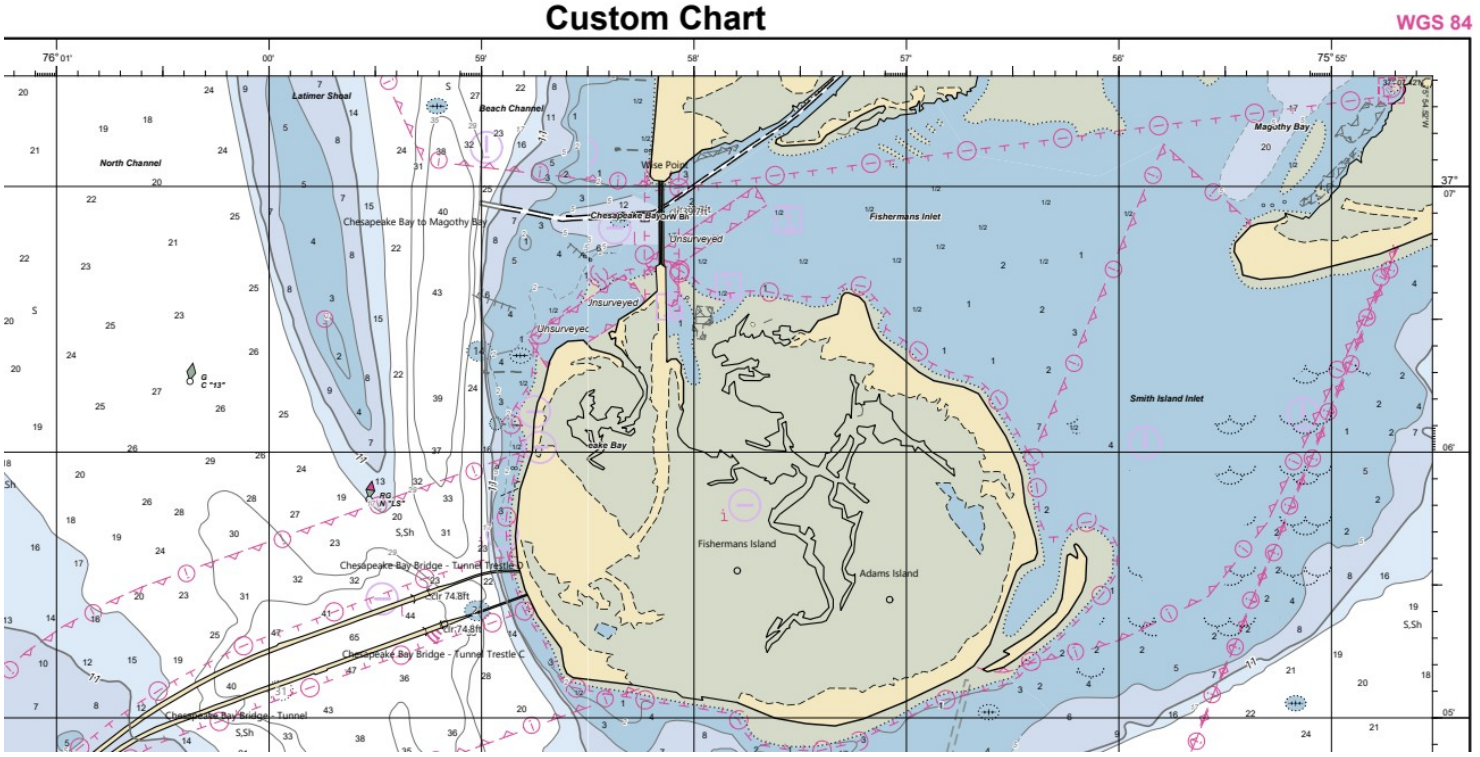
Different Base Maps for Different Purposes

Aeronautical map



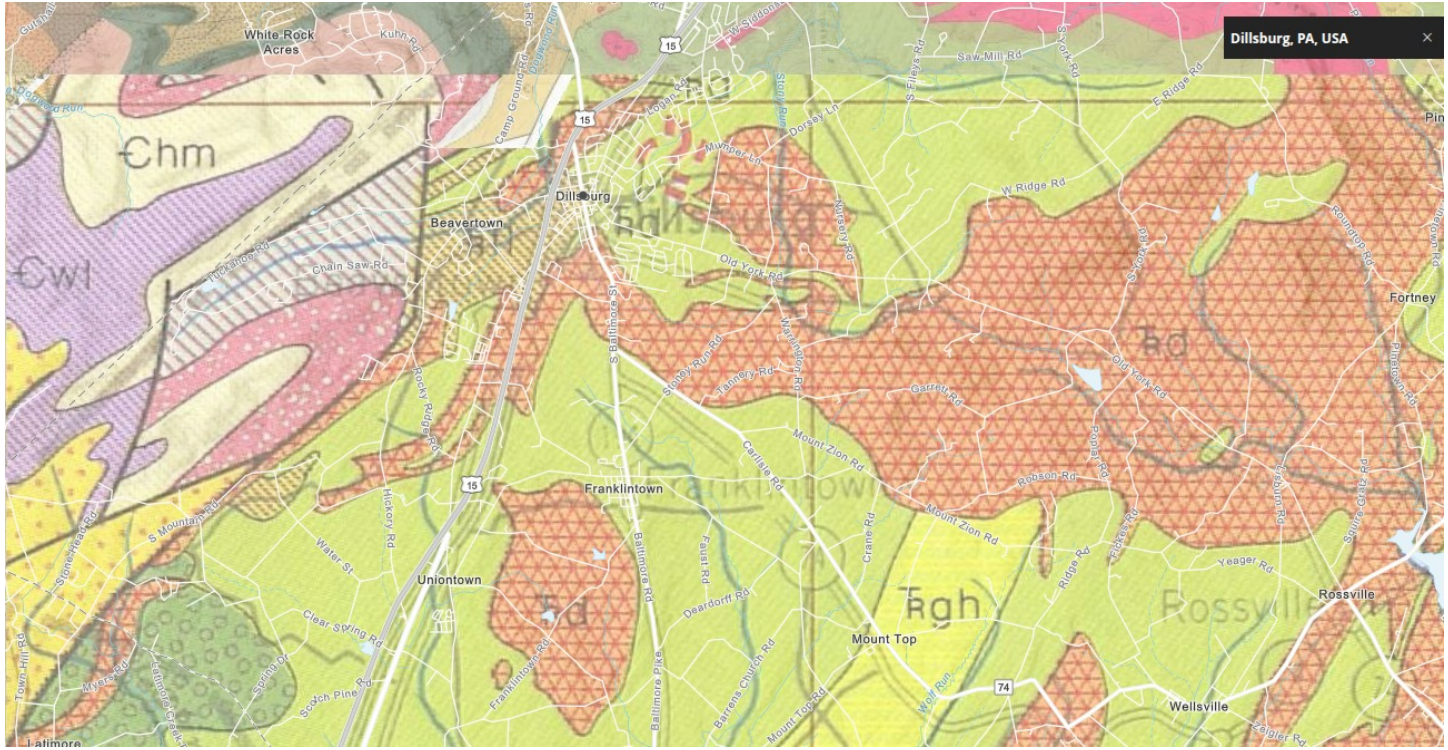
Different Base Maps for Different Purposes

Nautical map



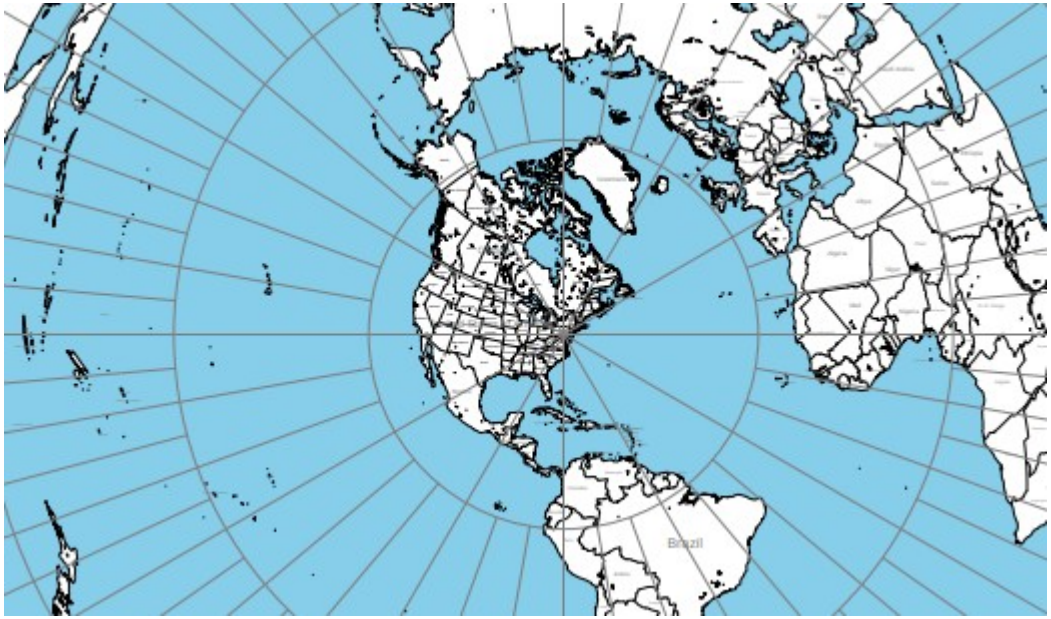
Different Base Maps for Different Purposes

Geological map



Different Base Maps for Different Purposes

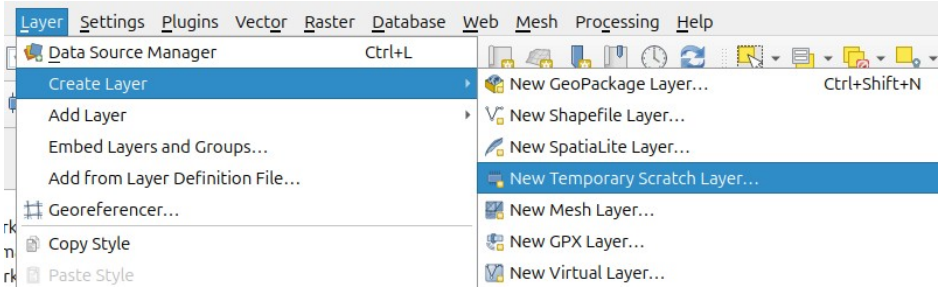
Azimuthal Projection Map https://ns6t.net/azimuth/

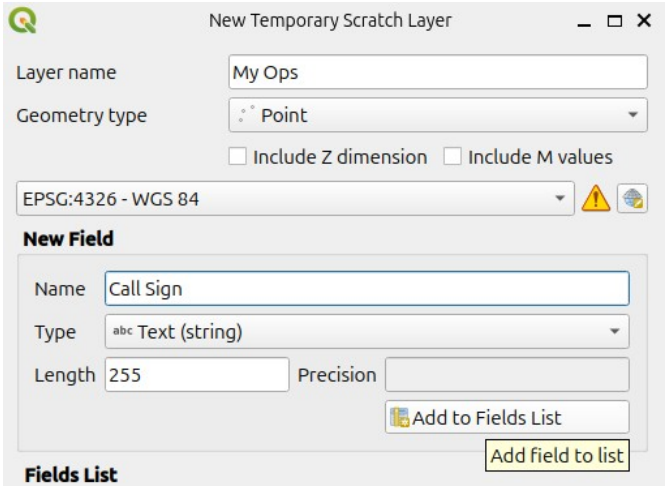


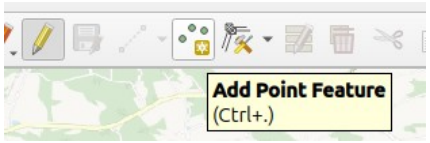
Typical Map Project Workflow

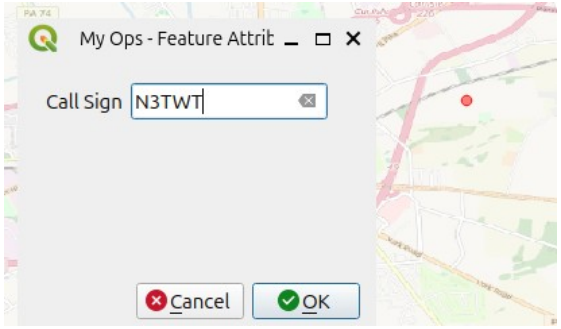
1	**Start a project**	New project → set CRS (or let it be set by first layer)
2	**Add a base map**	XYZ tiles (OSM) or local offline tiles / county imagery. Set transparency to remove visual “noise”
3	**Create point layer**	Stations with callsign, lat/lon, notes attributes
4	**Create line layer**	Contacts; attributes for signal strength, mode, notes
5	**Style & label**	Color lines by strength; label ends with callsigns
6	**Print / export**	Layout manager for PDF/print; or share the .qgz project

Create a Point Layer



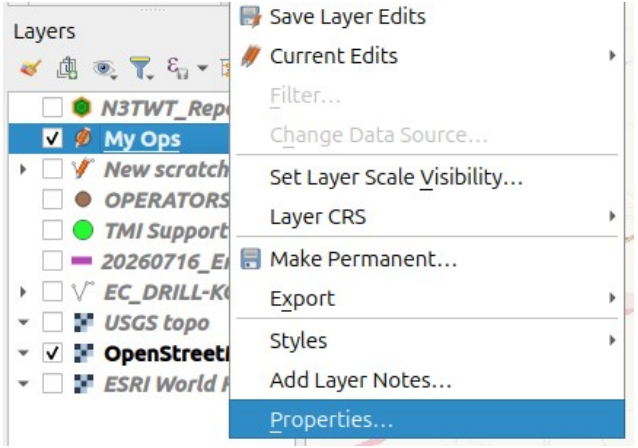




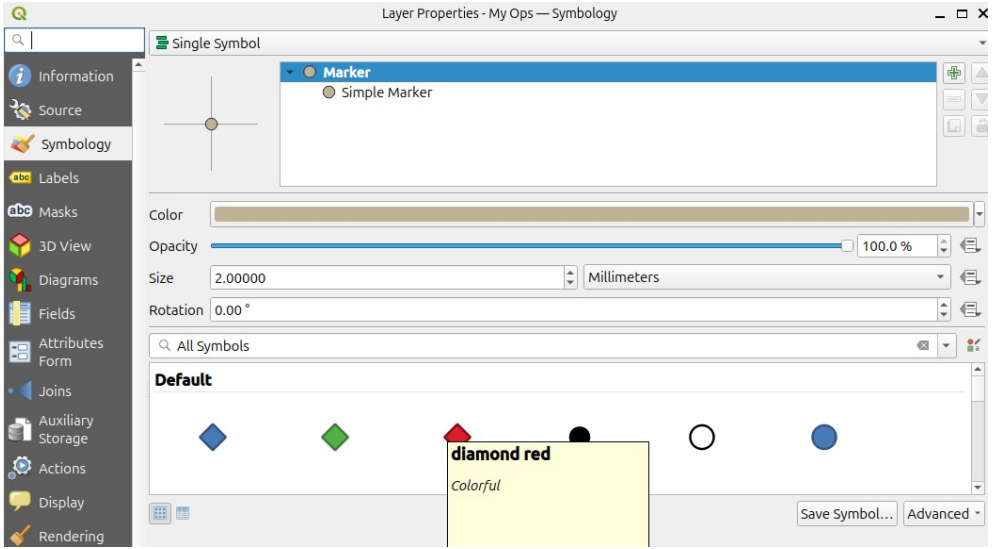


Create a Point Layer

Right click on the layer then Properties to change symbols



You can change the shape, color, icons, etc to make the points unique



Hands-on Focus: Points + Lines

Point Layer — Stations

- callsign (text)
- name / tactical (text)
- role (EOC, field, home…)
- notes

Line Layer — Contacts

- from_call / to_call
- signal (Strong / Fair / Weak)
- mode (FM, SSB, digital…)
- time / net
- notes (relay needed?)

Styling by Signal Strength

In Layer Properties → Symbology → Categorized (or Rule-based)
This would cause the lines to change color to match the value in Signal Strength



Strong

Solid green — clear copy, reliable



Fair

Amber — usable with effort / QSB



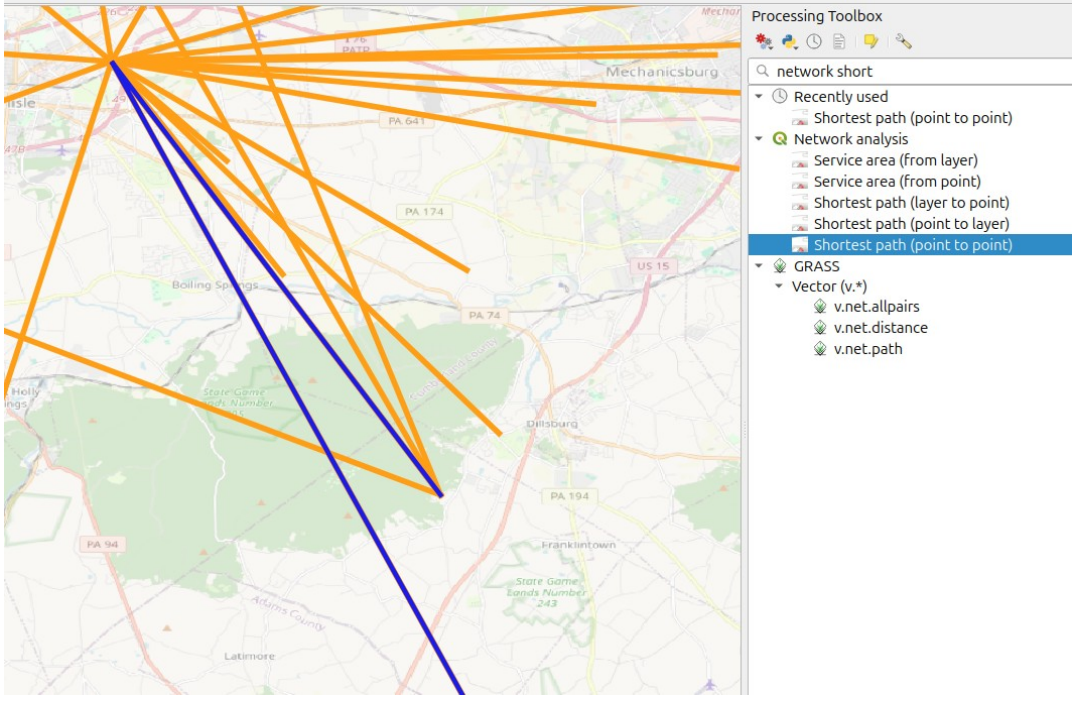
Weak

Red — marginal, needs relay

Shortest Path Analysis

QGIS can help answer questions about your data.
What is the best simplex path between 2 operators?

Two layers were merged and then the Shortest Path tool used to find the best way for me to contact Pat & Deb at their QTH



QGIS → CalTopo for Live Field Sharing

Do the careful layer work in QGIS, then push the results to CalTopo so field teams can see (and update) them in real time.

1. Export from QGIS

Reproject layer to WGS84 (EPSG:4326)

Right-click layer → Export → Save Features As…

Format: KML, GPX, or GeoJSON

- GPX – best for routes / tracks / waypoints
- KML – good all-around + attributes
- GeoJSON – preserves more structure

Have a “Name” field (KML) or “title” (GeoJSON) for labels (recommended)

Save the file

2. Import into CalTopo

Open (or create) your map at caltopo.com

Map Objects panel → Import

Select the KML / GPX / GeoJSON file

Check the features you want, click Import

Objects appear on the map & in the list

Then share the map (link or QR code)

so field teams see live updates

CERT Tip: Export windshield-survey routes as GPX tracks and resource locations as GPX waypoints or KML points for the cleanest results.

Useful External Data Sources

OpenStreetMap / XYZ tiles

Free basemap; can cache for offline use

County / State GIS portals

Parcels, roads, buildings, flood zones, EOCs

GPX from handhelds / APRS

Import track logs or waypoint exports

CSV / spreadsheet

Callsign + lat/lon list → Delimited Text layer

Repeater directories / FCC data

Public databases; convert to points

Your own digitizing

Click to add stations and draw contact lines

Practical Tips for EmComm Use

Offline readiness	Pre-download basemap tiles or use local GeoPackage / MBTiles so the map works without internet.
Print layouts	Use the Layout Manager to make clean 8.5×11 or 11×17 maps with legend, scale bar, and north arrow for the field.
Simple is better	One clear question per map. Too many layers = cluttered and hard to read under stress.
Share the project	A .qgz file + data folder can be handed to another operator who has QGIS.
Version & date	Always put date/time and “as of” on the map — situations change fast.

Other Resources

qgis.org
https://qgis.org — Download + official docs

QGIS Training Manual
https://docs.qgis.org/latest/en/docs/training_manual/

OpenStreetMap (OSM)
https://www.openstreetmap.org — Free basemap data & tiles

PASDA (PA Spatial Data)
https://www.pasda.psu.edu — Free PA GIS data (counties, EOCs, flood…)

CalTopo
https://caltopo.com — Real-time collaborative maps for field teams

QuickMapServices plugin
Plugins → Manage and Install Plugins → “QuickMapServices”

Cumberland / York County GIS
https://www.cumberlandcountypa.gov/3517/Map-Services · https://york-county-pa-gis-portal-yorkcountypa.hub.arcgis.com/

National GIS Day Nov. 18 https://www.harrisburgu.edu/events/central-pennsylvania-gis-day-2026/

Takeaways

1. QGIS lets you build custom maps that answer specific questions.
2. Points + lines + simple attributes (callsign, signal) already give huge value.
3. It complements (doesn’t replace) live tools like APRS or CalTopo.
4. Offline capability and print layouts make it useful in the field and for reports.
5. Start small: one net, one map, one clear picture of connectivity.

Quick Glossary

OSM / OpenStreetMap	Free community-built world map; common source of basemap tiles. openstreetmap.org
lat / lon	Latitude / Longitude – geographic coordinates (degrees). Standard for GPS and web maps.
WGS84	World Geodetic System 1984 (EPSG:4326). The global lat/lon datum used by GPS. ESRI equivalent: GCS_WGS_1984.
UTM	Universal Transverse Mercator – projected coordinate system (meters) good for local distance/area work.
CRS / Projection	Coordinate Reference System – how the curved Earth is represented on a flat map.
LIDAR	Light Detection and Ranging – laser scanning that produces detailed elevation / 3-D surface data.
Vector	Points, lines, or polygons that carry attributes (e.g., a station with a callsign).

Quick Glossary

Raster	Grid-based data such as aerial photos, elevation models, or LIDAR-derived surfaces.
Basemap	Background reference map (often OSM or imagery) under your own layers.
Layer	One dataset (or styled view) stacked in the map — like transparent sheets.
Attribute	Information stored with a feature: callsign, signal strength, notes, etc.
GeoPackage / Shapefile	Common file formats for storing and sharing vector GIS data.
WMS / XYZ tiles	Web services that deliver basemap images. XYZ is typical for OSM-style tiles.
QGIS Project (.qgz)	Project file that saves layers, styles, and layouts for reuse or sharing.

Questions?